



Report

Smart Night Vision Security System

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Abstract

The goal of this project was to create a practical and affordable infrared night vision camera capable of operating effectively in both total darkness and light. Using infrared (IR) technology, the system was designed to automatically adjust its visibility mode based on ambient light levels. This was achieved by combining an ESP32-CAM module, which recorded and streamed video over Wi-Fi to a laptop acting as the display, with an Arduino UNO connected to a light-dependent resistor (LDR) to detect light intensity. When light was detected, a green LED was activated; in darkness, the Arduino triggered three IR LEDs, which emitted infrared light invisible to the human eye but visible to the ESP32-CAM, allowing for clear imaging at night. The prototype was enclosed in a box designed to resemble a surveillance camera, adding both realism and practicality. The system delivered high-resolution video with minimal delay and performed automatic light detection and IR switching with high accuracy. Overall, the project successfully demonstrated the integration of sensor technology, infrared optics, and wireless video transmission, with strong potential applications in surveillance, security, and robotics for low-light environments.

Introduction

Human vision depends heavily on visible light, and in low-light or dark conditions, our ability to see is greatly reduced. This poses challenges in areas such as security surveillance, robotics, military operations, wildlife monitoring, and personal safety. To address these issues, this project focused on creating a low-cost, practical night vision system using accessible components. Many commercial night vision systems are expensive, bulky, and difficult to set up, making them unsuitable for personal or small-scale use. Our solution aimed at overcoming these limitations by offering a portable, wireless, and easy-to-use alternative that can be quickly deployed in various environments without the need for technical expertise.

The system uses an ESP32-CAM module for video capture and wireless transmission, and an Arduino UNO to control lighting based on ambient conditions. A Light Dependent Resistor (LDR) detects light levels: if light is present, a green LED turns on; in darkness, infrared LEDs are activated, allowing the ESP32-CAM to continue streaming clear video using IR light invisible to the human eye. The feed is transmitted to a laptop over Wi-Fi for real-time monitoring. All components are housed in a custom enclosure designed to resemble a professional surveillance camera. The system was tested in different environmental conditions, including temperature extremes and wind, and remained stable throughout. Its automatic light adjustment and wireless design make it intuitive and suitable for real-world use, proving that effective night vision technology can be both affordable and user-friendly.

Methodology

This project focused on designing and implementing a low-cost, functional night vision camera system capable of automatically adapting to changing light conditions. The process included selecting hardware, programming microcontrollers, assembling and testing the system, and improving the prototype for real-world use. Below is a detailed explanation of the tools, techniques, and materials used.

1. Hardware Components and Tools

The system was built using the following components:

- **ESP32-CAM module (AI Thinker)** – Used for capturing and streaming video wirelessly.
- **FTDI Programmer** – Used for Uploading codes on ESP32-CAM
- **Arduino UNO** – Used for reading ambient light levels and controlling LED outputs.
- **ULTRASONIC SENSOR (HC-SR04)** – A sensor to detect people approaching the camera
- **Buzzer** – Makes sound when people approaching the camera
- **Light Dependent Resistor (LDR)** – A sensor to detect surrounding light intensity.
- **Yellow LED** – Activated in well-lit environments to indicate visibility.
- **Infrared LED** – Provide illumination in dark environments, visible only to the IR-sensitive ESP32-CAM.
- **Breadboard and jumper wires** – For prototyping and temporary circuit assembly.
- **USB cables and a laptop** – For powering and programming both microcontrollers.



ESP32-CAM



FTDI



ARDUINO UNO



ULTRASONIC



LDR KY-018



Yellow LED



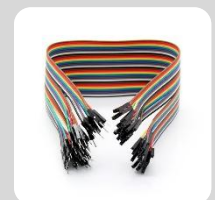
IR LED



BUZZER



BREADBOARD

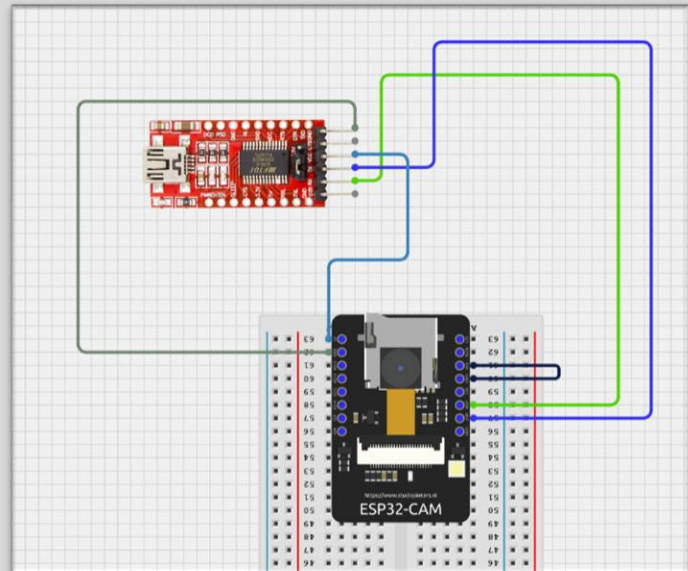


Jumper wires

2. Programming and Configuration

- **ESP32-CAM Configuration:**

- The ESP32-CAM was programmed using the **Arduino IDE** with the **ESP32 board package** installed.
- The camera module selected was "**AI Thinker ESP32-CAM**".
- The board was connected via **COM7**, and an upload speed rate of **115200** was used to ensure stable communication.
- The firmware enabled the ESP32-CAM to connect to 1



a Wi-Fi network (same as the laptop's), allowing video to be streamed in real-time to a web browser on the laptop. This effectively turned the laptop into a display screen.

FTDI PROGRAMMER	ESP32-CAM
VCC	5V
GND	GND
TX	U0R
RX	U0T

Table 1

Arduino UNO Configuration:

programmed using **Arduino IDE**.

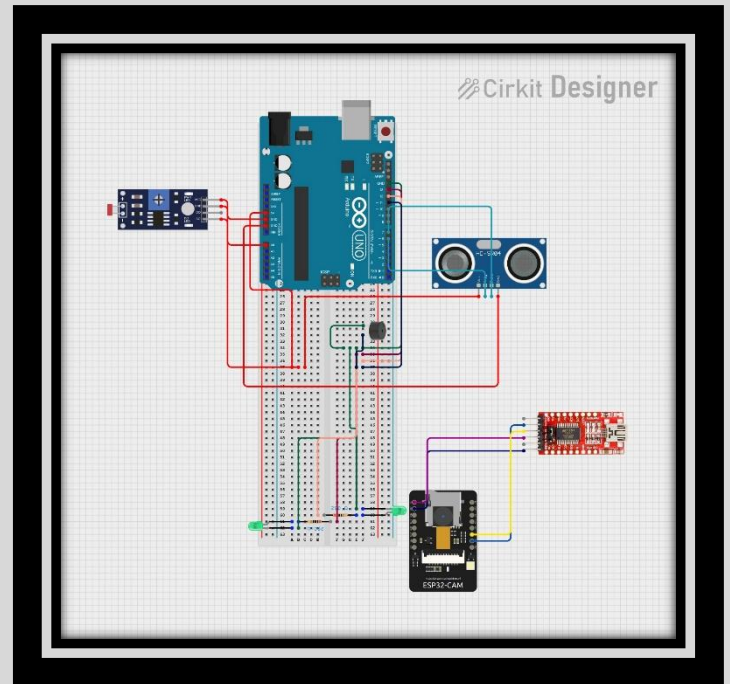
Three main tasks were implemented:

1. Light detection logic with LDR:

- If the analog reading > 500 (bright light), the green LED is activated.
- If < 500 (dark), the IR LEDs are turned on to assist night vision.

2. Ultrasonic sensing logic:

- Continuously reads distance using the HC-SR04 sensor.
- If an object is detected at **less than 10 cm**, the **buzzer** is **triggered** as a warning alert.



2

3. All logic used standard **digital/analog read-write functions** from the Arduino library.

Ultrasonic Sensor (HC-SR04)	Arduino UNO
VCC	5V
GND	GND
Trig	Pin 9
Echo	Pin 10

Table 2

LDR KY-018	Arduino UNO
VCC	5V
GND	GND
D0	No connection
A0	A0

Table 3

Buzzer	Arduino UNO
Anode +ve (Long)	Pin 11
Cathode -ve (Short)	GND

Table 4

Yellow LED	Arduino UNO
Anode +ve (Long)	Pin 12
Cathode -ve (Short)	GND

Table 5

IR LED	Arduino UNO
Anode +ve (Long)	Pin 13
Cathode -ve (Short)	GND

Table 6

3. Power Supply and Circuit Assembly

- Both the **ESP32-CAM** and **Arduino UNO** were powered via **USB ports connected to the laptop**, which supplied a regulated **5V DC output**. This made the system portable and easy to manage without the need for external batteries or voltage regulators.
- The prototype circuit was first assembled on a **breadboard** using **jumper wires**, allowing for easy adjustments and debugging during the testing phase. This also facilitated component swapping and layout optimization.

4. Physical Enclosure

- The hardware was enclosed in a **custom-built box**, constructed from **cardboard**, and designed to resemble a traditional **surveillance camera**.
- This not only provided basic protection and mounting convenience but also enhanced the professional appearance of the device, making it suitable for real-world deployment and demonstration purposes.

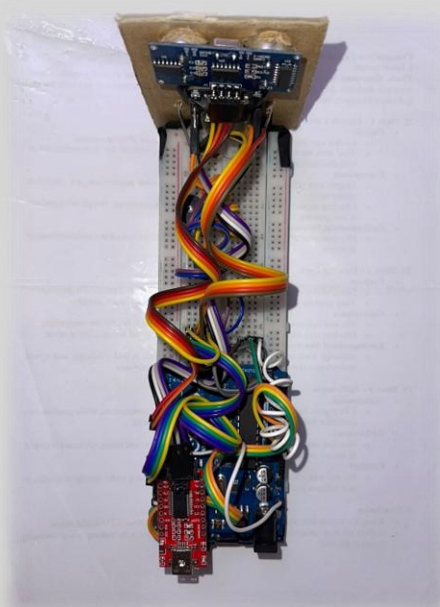
5. Testing and Validation

- The functionality of the light detection system was verified through **controlled experiments**:
 - To simulate darkness, the **LDR sensor was placed inside a closed dark box**. As expected, the **IR LEDs turned on**, and the ESP32-CAM was able to capture and stream video clearly, thanks to the infrared illumination.

- When the LDR was exposed to ambient light, the **green LED activated**, and the IR LEDs remained off, showing the system's ability to switch modes automatically based on lighting conditions.
- The ultrasonic sensor was tested by placing objects at various distances. When an object was detected within 10 cm, the buzzer was successfully triggered, providing a clear and immediate warning signal. This confirmed the sensor's effectiveness in detecting nearby obstacles.
- Additional environmental tests were conducted in **varied temperature and wind conditions**, both indoors and outdoors, to assess the durability and reliability of the system. The prototype functioned correctly without being affected by external factors, confirming its robustness and suitability for real applications.

6. System Integration and Interaction

- The two microcontrollers operated independently but in synchronization:
 - The Arduino UNO handled **sensing and LED control** logic.
 - The ESP32-CAM focused on **image acquisition and transmission**.
- This modular design allowed each component to focus on its role, enhancing overall system efficiency, performance, and maintainability.
- The **wireless feature** enabled flexible positioning of the camera, as no direct cable connection between the camera and display was needed.



conclusion

This project successfully achieved its core objective of creating a cost-effective, portable, and reliable night vision camera system that can adapt automatically to varying light conditions and provide real-time visual monitoring and proximity alerts. By combining the ESP32-CAM for wireless video streaming with the Arduino UNO for light and distance sensing, the system effectively monitored environmental changes and responded accordingly. In bright conditions, a green LED was activated, while in dark environments, infrared LEDs turned on to enable invisible illumination, allowing the ESP32-CAM to maintain clear video transmission. The integration of an ultrasonic sensor added a valuable layer of functionality, enabling the detection of objects within a 20 cm range.



When an object was detected at close range, the buzzer was triggered, producing an audible warning that enhances safety and user awareness, especially useful in security or navigation applications. The entire system was tested under different conditions, including varying temperatures and airflow, and proved to be stable, durable, and consistently responsive. Its modular design, use of readily available components, and intuitive user interface through a browser-based video feed make it practical and scalable. Key insights include the effectiveness of infrared light for night vision, the advantages of modular microcontroller integration, and the value of combining multiple sensors for enhanced functionality. In conclusion, the project met all its goals and demonstrated strong potential for applications in surveillance, home automation, robotics, and safety systems. Future improvements could involve weatherproofing the enclosure, increasing wireless range, enhancing power efficiency, and incorporating advanced features such as cloud storage, mobile app integration, or AI-based motion detection for smarter monitoring.



Recommendations / Future Work

Battery Power Supply

Shifting from USB power to a rechargeable battery pack would make the device fully portable and independent of laptop power, expanding its application to remote and off-grid environments.

Motion Detection Integration

Adding a Passive Infrared (PIR) sensor or using built-in ESP32 camera motion detection libraries could enable the device to activate video streaming only when motion is detected, saving power and bandwidth.

Cloud or Local Storage

Implementing data logging by saving video footage to an SD card or uploading it to a cloud service would allow for surveillance playback and record-keeping.

Mobile App Interface

Creating a simple mobile application to view the camera feed and control settings would enhance user experience and make the system more accessible.

Extended Range and Signal Stability

Enhancing Wi-Fi performance or incorporating mesh networking capabilities would improve connectivity, especially in large or obstructed areas.

AI-Based Image Enhancement

Future iterations could integrate AI tools to enhance low-light image quality, detect intruders, or distinguish between humans and animals.

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